



GROUP ASSESSMENT COVERSHEET

Trimester 2620

Section A: Submission Details

Programme	:	Bachelor of Computer Science
Course Code & Name	:	LDCW6123 - FUNDAMENTALS OF DIGITAL COMPETENCE FOR PROGRAMMER
Course Lecturer(s)	:	Mr. V Segaran A/L Veeraya
Submission Title	:	Group Project - Spotify and Physical Music Clayton Model and Interactive Program
Deadline	:	Day 2 Month 10 Year 2026 Time 2359
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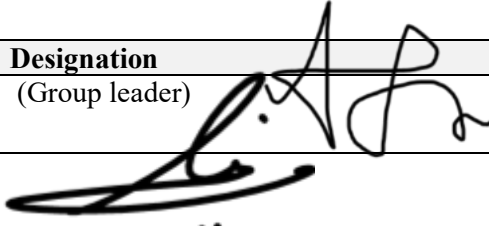
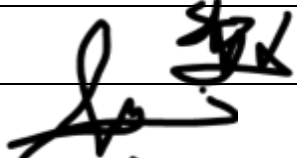
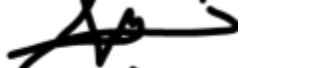
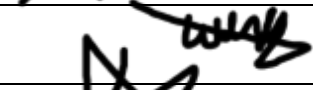
Section B: Academic Integrity

Tick (✓) each box below if you agree:

☒	I have read and understood the policy on Plagiarism in University Rules & Regulations.
☒	This submission is my own, unless indicated with proper referencing.
☒	This submission has not been previously submitted or published.
☒	This submission follows the requirements stated in the course.

Section C: Submission Details

Group 13

Student ID(s)	Student Name(s)	Designation (Group leader)
261UC240LY	Aziel Tan Zheng Chuan	
261UC240PJ	See Wing Kit	
261UC26145	Soo Kian Rong	
261UC2406W	Vincent Lock Chun Kit	
253UC255P1	Wong Kee Yuan	
262UC250T8	Ho Ming Hao	

Declaration by group leader

I hereby declare that all group members' names are correctly included in the above section. I hold a copy of this assignment which I can produce if the original is lost or damaged. I certify that no part of this assignment has been copied from any other student's work or from any other source except where due acknowledgement is made in the assignment.

Group leader's signature :  _____

Group leader's name : Aziel Tan Zheng Chuan

Group leader's student ID : 261UC240LY

Date : 22/09/2026

Group member declaration

For the purpose of completing this assignment, I have performed the following tasks:

1. Coordinated the compilation and organisation of the group report and supporting materials.
2. Reviewed the consistency between the report, presentation slides, and demonstration script.
3. Reviewed the program-output evidence and the accompanying explanations included in the report.
4. Prepared a demonstration scenario showing a recommendation that matched all selected preferences.
5. Prepared a second scenario showing a partial match and an explanation that listed only the preferences actually matched.
6. Demonstrated invalid-input rejection, recovery after valid input, repeated searches, and normal program termination.
7. Prepared and rehearsed the presentation script and live demonstration sequence for Slides 9–10.
8. Presented the program demonstration and clarified that the catalogue contained fictional tracks and artists.

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature : _____

Group member's name : Aziel Tan Zheng Chuan

Group member's ID : 261UC240LY

Date : 02/10/2026

Group member declaration

For the purpose of completing this assignment, I have performed the following tasks:

1. Contributed to the report sections covering testing, limitations, and the conclusion.
2. Reviewed the recorded test results and their correspondence with the stated test coverage.
3. Explained the coverage of preference combinations, invalid inputs, end-of-input cases, and additional program behaviour.
4. Distinguished successful functional tests from evidence of recommendation usefulness to real listeners.
5. Reviewed the development-history evidence and its explanation in the report.
6. Reviewed the conclusion to ensure that it acknowledged the conceptual graph, other market influences, and the continued presence of physical music.
7. Prepared and rehearsed the presentation script for Slides 11–12.
8. Presented the testing evidence, discussed the project's limitations, and delivered the closing conclusion.

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature : _____

Group member's name : See Wing Kit

Group member's ID : 261UC240PJ

Date : 02/10/2026

Group member declaration

For the purpose of completing this assignment, I have performed the following tasks:

1. Contributed to the report sections describing Spotify's historical development and its impact on the music market.
2. Reviewed Spotify's service milestones and organised the chronology used in the report and presentation.
3. Reviewed the supporting sources for the historical dates and music-industry figures.
4. Distinguished industry-wide streaming revenue figures from Spotify-specific information to avoid misleading comparisons.
5. Explained how streaming changed music access, distribution, and subscription-based and advertising-supported payment models.
6. Discussed the continued presence of CDs and other physical music formats, acknowledging that Spotify was not the sole cause of market changes.
7. Prepared and rehearsed the presentation script for Slides 5–6, ensuring consistency with the report.
8. Presented Spotify's milestones, the market evidence, and the implications of streaming for physical music.

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature

:  _____

Group member's name : Soo Kian Rong

Group member's ID : 261UC26145

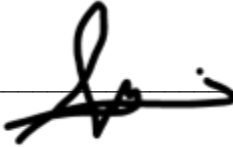
Date : 02/10/2026

Group member declaration

For the purpose of completing this assignment, I have performed the following tasks:

1. Contributed to the introduction and background of the Spotify and CD comparison in the report.
2. Helped define the scope of the case study and identify CDs as the established option being compared with Spotify.
3. Explained Spotify's alternative value of on-demand catalogue access and music discovery.
4. Clarified the use of delivered digital audio fidelity as the conventional performance measure in the comparison.
5. Reviewed the introduction to ensure consistency with the poster and the subsequent model analysis.
6. Explained the connection between the innovation case and the C++ Music Discovery Assistant.
7. Prepared and rehearsed the presentation script for Slides 1–2.
8. Presented the introduction, case scope, and main question during the group presentation.

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature : 

Group member's name : Vincent Lock Chun Kit

Group member's ID : 261UC2406W

Date : 02/10/2026

Group member declaration

For the purpose of completing this assignment, I have performed the following tasks:

1. Contributed to the report's description of the C++ Music Discovery Assistant and its purpose.
2. Reviewed the Track structure and the fictional catalogue used by the program.
3. Explained the user inputs for genre, mood, and preferred track duration.
4. Reviewed the genre-filtering process, preference-scoring rules, and handling of equal scores.
5. Explained the whole-line input validation and rejection of invalid menu entries.
6. Checked that the report's descriptions of recommendation explanations and fallback behaviour corresponded to the source code.
7. Prepared and rehearsed the presentation script for Slides 7–8.
8. Presented the program design and explained its relationship to digital music discovery.

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature : 

Group member's name : Wong Kee Yuan

Group member's ID : 253UC255P1

Date : 02/10/2026

Group member declaration

For the purpose of completing this assignment, I have performed the following tasks:

1. Contributed to the report section applying Clayton Christensen's disruptive innovation model to Spotify and CDs.
2. Reviewed the established and emerging technology trajectories shown in the graph.
3. Prepared explanations of the customer-demand levels and their relationship to the selected audio-fidelity measure.
4. Explained why the CD trajectory remains level under the fixed digital-format comparison.
5. Clarified the distinction between audio fidelity and the alternative benefits of immediate access and music discovery.
6. Reviewed the graph labels and report explanations to ensure that conceptual trajectories were not presented as measured data.
7. Prepared and rehearsed the presentation script for Slides 3–4.
8. Presented the graph interpretation and explained the limitations of its conceptual trajectories.

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature : 

Group member's name : Ho Ming Hao

Group member's ID : 262UC250T8

Date : 02/10/2026



LDCW6123 - FUNDAMENTALS OF DIGITAL COMPETENCE FOR PROGRAMMER

Group Project

Spotify and Physical Music Clayton Model and Interactive Program

Group: 13

Lecture's Name: Mr. V Segaran A/L Veeraya

No	Student ID	Student name
1	261UC240LY	Aziel Tan Zheng Chuan
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Faculty of Computer and Informatics

Semester 2620

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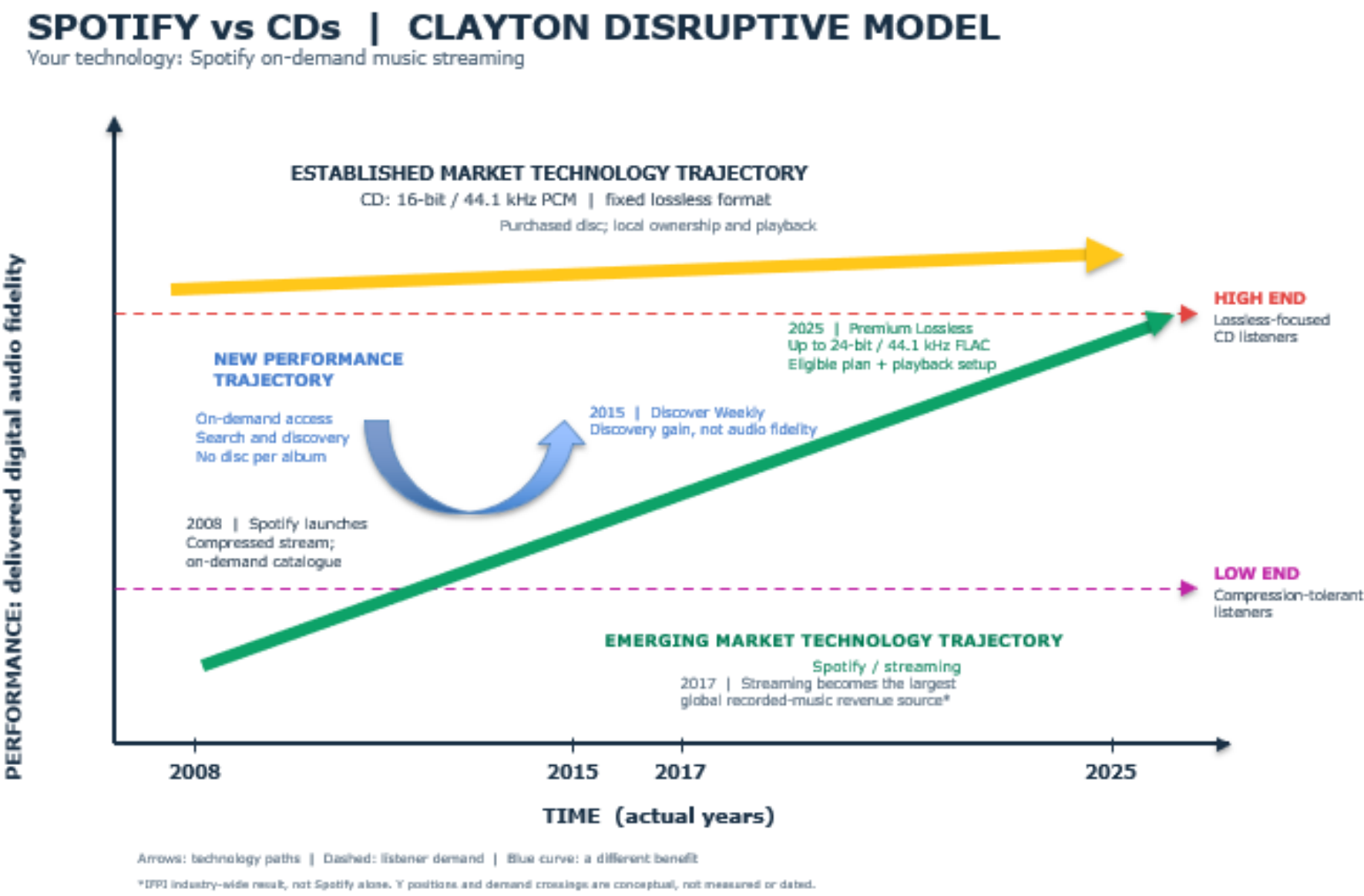
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FROM OWNERSHIP TO ACCESS

Spotify vs CDs / Clayton Christensen disruptive innovation model / LDCW6123 Group 13



ESTABLISHED MARKET

A purchased CD provides a lossless 16-bit/44.1 kHz digital signal and lasting possession of that disc (Sony, n.d.).

EMERGING ENTRANT

Spotify launched in 2008 with access to licensed music. Early compressed streaming traded conventional signal fidelity for immediacy (Spotify, n.d.).

DIFFERENT VALUE

Search and discover music on demand without buying each CD. Mobile access arrived in 2009; Discover Weekly followed in 2015 (Spotify, n.d.).

CUSTOMERS

High-end: lossless-focused CD listeners. Lower conventional demand: people willing to accept compressed playback for access. These are needs, not income groups.

MODEL LIMIT

The level CD path reflects a fixed format. The sharp 2025 green turn denotes a new Premium option, not all streams. The 2017 industry revenue lead is not an audio-quality crossing.

2008

Spotify launches in six European markets

2009

Mobile access

2015

Discover Weekly

2017

Streaming leads global recorded-music revenue*

2025

Premium Lossless rollout; physical formats persist

SELECTED REFERENCES

Christensen Institute. (n.d.). Disruptive innovation theory.
<https://www.christenseninstitute.org/theory/disruptive-innovation/>

International Federation of the Phonographic Industry. (2018). IFPI global music report 2018. <https://www.ifpi.org/ifpi-global-music-report-2018/>

International Federation of the Phonographic Industry. (2026, March 18). Global music report 2026: Global recorded music revenues grow 6.4% as record companies drive

innovation. <https://www.ifpi.org/global-music-report-2026-global-recorded-music-revenues-grow-6-4-as-record-companies-drive-innovation/> *IFPI (2018) reports 38.4% of

global industry revenue in 2017. Conceptual graph: vertical positions are not numerical measurements.

Sony. (n.d.). A quick guide in understanding digital audio. <https://www.sony.com/electronics/support/articles/00165079>

Spotify. (2025, September 10). Lossless listening arrives on Spotify Premium with a richer, more detailed listening experience. <https://newsroom.spotify.com/2025-09-10/lossless-listening-arrives-on-spotify-premium-with-a-richer-more-detailed-listening-experience/>

Spotify. (n.d.). Spotify: A visual history. <https://newsroom.spotify.com/spotify-timeline/>

Part 1 Model application

Scope and model

This analysis focuses on purchases of compact discs (CDs) as the incumbent offering in the physical recorded-music market. The entrant is Spotify, a licensed, on-demand streaming service launched in six European countries in October 2008 (Spotify, n.d.). Christensen's model examines how entrants establish a foothold among overserved customers or nonconsumers, offer greater affordability or accessibility, and improve enough to challenge established providers (Christensen Institute, n.d.). This poster assesses whether Spotify fits a low-end disruption pathway rather than assuming that a new digital service is necessarily disruptive.

Foothold and customers

For this model, the more demanding customer is a listener who values owning a physical recording and preserving CD-quality audio. A listener with lower fidelity requirements may accept lossy streaming in exchange for easy search and access to a large licensed catalogue. "Low-end" refers here to customers whose needs may be overserved by the existing offering, rather than their income. Spotify's free and paid options suggest a possible foothold among these listeners. However, their preferences alone do not establish low-end disruption: the argument also requires evidence of a lower-cost business model and limited incumbent incentives to compete for them. Other digital alternatives were already available, so this customer mapping remains a proposed explanation.

Performance and customer mapping

The graph's vertical axis represents digital audio fidelity relative to a lossless source, rather than the overall value of the listening experience. Standard CD audio uses 16-bit/44.1 kHz encoding, while Spotify's earlier streaming relied on lossy compression. In September 2025, Spotify began rolling out lossless audio within Premium, supporting FLAC at up to 24-bit/44.1 kHz (Sony, n.d.; Spotify, 2025).

The yellow CD arrow is horizontal because the standard CD format remains fixed. The green Spotify line starts below the CD benchmark and reaches the lossless region to represent the 2025 rollout. Its earlier slope and crossing of the "good enough" threshold are conceptual assumptions, not measured changes or established dates. The graph illustrates a possible performance trajectory; it does not demonstrate an audible quality gap or superiority over CDs. The lossless specification also does not mean that every track, setting or playback connection delivers lossless audio.

A second, distinct value

The blue curved callout represents Spotify's appeal beyond audio fidelity; it is not another vertical scale. Spotify offered on-demand access to a large music catalogue and quick search, later adding personalised discovery features such as Discover Weekly, launched in 2015 (Spotify, 2025). Listeners could explore music without buying each album or storing physical discs. This convenience could make streaming attractive to users willing to accept lower digital audio fidelity in exchange for easier access.

Technology and business models

Licensing and digital distribution allowed one service to offer on-demand tunes across devices; mobile access was added in 2009, according to Spotify's timeline. The free tier and premium Premium have varied access and quality options. The commercial relationship is not the same as selling a CD that you own forever. These distinctions tie the technology, business model and the trade-off of the audience (Spotify, n.d.; Christensen Institute, n.d.).

Part 1 Historical development and impact

Verified timeline

Spotify launched in October 2008, introduced its mobile service in September 2009 and added a free mobile experience in 2013 (Spotify, n.d.). Discover Weekly followed in 2015 (Spotify, 2025). By 2017, streaming had become the largest source of global recorded-music revenue, accounting for 38.4% of the total. This figure covers streaming across the industry, rather than Spotify's market share (IFPI, 2018). In September 2025, Spotify began rolling out lossless audio to Premium subscribers in selected markets (Spotify, 2025).

Interpreting the performance path

On the graph's narrow measure of digital fidelity, Spotify's early lossy streaming sits below the CD benchmark. The lower customer-demand line represents listeners for whom streaming was already "good enough" before Spotify introduced lossless audio. This is a conceptual customer mapping, rather than a measured threshold.

Streaming became the industry's largest revenue source in 2017, well before Spotify's lossless rollout. This suggests that access and convenience mattered alongside fidelity, although the revenue figures do not establish why individual listeners chose streaming. The green endpoint represents Spotify's later lossless upgrade for Premium subscribers; it does not imply that every track or playback connection delivers lossless audio (IFPI, 2018; Spotify, 2025).

Access, revenue and social impact

Streaming expanded access to music by allowing listeners to search a licensed catalogue without purchasing each album. Services generate revenue through subscriptions and advertising. Mobile listening and recommendations can make music easier to explore, particularly for people with limited storage space or access to physical music shops.

In 2025, streaming accounted for 69.6% of global recorded-music revenue, while revenue from physical formats increased by 8.0% (IFPI, 2026). These figures show streaming's dominance alongside continuing demand for physical music. They describe the industry as a whole and do not establish how much of the change was caused by Spotify.

Part 1 Tradeoffs and critical interpretation

Costs and limits

A purchased CD can be kept and played on suitable equipment without an ongoing subscription. Streaming access depends on the available catalogue, supported devices and service terms. Spotify Premium offers downloads for offline listening, but these remain tied to the service rather than providing ownership of a physical copy (Spotify, 2019).

Audio quality also depends on the encoding, subscription plan, device and playback connection. Spotify warns that Bluetooth can require compression even when the source is lossless (Spotify, 2025). Meanwhile, continued demand for vinyl and other physical formats shows that some listeners still value tangible music collections (IFPI, 2026).

Critical interpretation

Spotify demonstrates how listeners might accept lower digital audio fidelity in exchange for easier access to music. However, classifying it as low-end disruption under Christensen's model requires more evidence. Digital downloads and other services already existed, and streaming's revenue growth in 2017 does not establish the foothold or incumbent response required by the theory.

Established record labels also licensed their catalogues to Spotify, complicating a simple account of entrants replacing incumbents. The graph therefore presents a reasoned application of the model rather than measured market share, audio-test results or proof of causation. Its horizontal CD line represents the fixed technical specification of standard CD audio.

Connection to Part 2

Part 2 explores music discovery through an offline C++ Music Discovery Assistant. It matches users' stated preferences to a small catalogue of fictional artists using explicit rules. The project demonstrates a simple approach to preference-based discovery without using Spotify accounts, Spotify data or Spotify's recommendation algorithms.

Part 1 References

Christensen Institute. (n.d.). *Disruptive innovation theory*.

<https://www.christenseninstitute.org/theory/disruptive-innovation/>

International Federation of the Phonographic Industry. (2018). *IFPI global music report 2018*.

<https://www.ifpi.org/ifpi-global-music-report-2018/>

International Federation of the Phonographic Industry. (2026, March 18). *Global music report 2026:*

Global recorded music revenues grow 6.4% as record companies drive innovation.

<https://www.ifpi.org/global-music-report-2026-global-recorded-music-revenues-grow-6-4-as-recordcompanies-drive-innovation/>

Multimedia University. (n.d.). *Digital convergence and emerging innovative problem solving through:*

Topic 6A [Lecture slides]. LDCW6123.

Part 1 References continued

Sony. (n.d.). *A quick guide in understanding digital audio.*

<https://www.sony.com/electronics/support/articles/00165079>

Spotify. (2019, May 24). *How to download and listen to music and podcasts - offline and on the go.*

<https://newsroom.spotify.com/2019-05-24/how-to-download-and-listen-to-music-and-podcastsoffline-and-on-the-go/>

Spotify. (2025, September 10). *Lossless listening arrives on Spotify Premium with a richer, more detailed listening experience.*

<https://newsroom.spotify.com/2025-09-10/lossless-listening-arrives-on-spotify-premium-with-aricher-more-detailed-listening-experience/>

Spotify. (n.d.). *Spotify: A visual history.*

<https://newsroom.spotify.com/spotify-timeline/>

Part 2 Programmed design

Purpose and integration

The Music Discovery Assistant helps a user explore 12 fictional tracks by genre, mood and preferred duration. It shows how catalogue discovery can work in access-based music services, without connecting to Spotify or any external API.

Input and output

Genres are Pop, Electronic and Acoustic. Moods are Calm, Energetic and Reflective. A length option asks for up to four minutes or over four minutes. Output shows the fictional track title, artist, duration, description and the matched preferences. A repeat prompt allows another search.

Transparent selection rule

Genre is a required filter. A matching mood adds two points, and a matching length adds one. The first track in catalogue order wins an equal score. Mood therefore outranks length. The explanation prints only matches actually achieved; an absent genre in a future edited catalogue has a safe fallback.

Implementation and validation

C++17 uses a Track struct, a vector catalogue, loops and conditional scoring. readChoice consumes the entire line and uses istringstream to accept only one bounded integer with optional surrounding whitespace. It rejects text, decimals, trailing characters, blank input, overflow and out-of-range choices. End-of-input exits cleanly. Thirty-five black-box checks exercise 18 preference combinations and the key invalid-input, EOF, repeat and fallback cases.

Limits

Tracks and artists are invented, the catalogue is small, and weights were chosen for a classroom demonstration. No machine learning, personalized history, account, persistence, streaming, or usability study is implemented. Tests establish deterministic behaviors, not satisfaction with the selected music.

Build: g++ -std=c++17 -Wall -Wextra -pedantic src/main.cpp -o music assistant

Part 2 Input validation source

The image is generated from the actual C++ file. readChoice validates each whole line before accepting a bounded integer. Full source is included in src/main.cpp.

```
33 int readChoice(const std::string& prompt, int minimum, int maximum) {
34     while (true) {
35         std::cout << prompt;
36         std::string line;
37         if (!std::getline(std::cin, line)) {
38             std::cout << "\nInput ended. Goodbye.\n";
39             std::exit(0);
40         }
41         std::istringstream input(line);
42         int choice = 0;
43         char extra = '\0';
44         if (input >> choice && !(input >> extra)
45             && choice >= minimum && choice <= maximum) {
46             return choice;
47         }
48         std::cout << "Please enter a whole number from " << minimum
49             << " to " << maximum << ".\n";
50     }
51 }
```

Figure 1: Actual readChoice source with line numbers.

Part 2 Recommendation source

A required genre filter, mood weight of two and length weight of one determine each result. Equal scores retain the first track; a missing-genre guard prevents a null deference.

```
67         int bestScore = -1;
68         const Track* best = nullptr;
69         for (const Track& track : catalogue) {
70             if (track.genre != genres[genre - 1]) continue;
71             int score = 0;
72             if (track.mood == moods[mood - 1]) score += 2;
73             if ((track.minutes <= 4) == (length == 1)) score += 1;
74             // On equal scores retain the first track in catalogue order.
75             if (score > bestScore) {
76                 best = &track;
77                 bestScore = score;
78             }
79         }
80         // Guard against a future catalogue without the requested genre.
81         if (best == nullptr) {
82             std::cout << "No tracks are available in this genre.\n";
83             again = readChoice("Find another? 1 Yes 2 No: ", 1, 2) == 1;
84             continue;
85         }
```

Figure 2: Actual selection code and fallback.

Part 2 Programmed output and validation

A genuine terminal session recommends Slow Orbit, then City Lanterns after a second search. The explanation does not claim a Reflective mood match for City Lanterns.

```
MUSIC DISCOVERY ASSISTANT
An offline, rule-based demonstration of digital music discovery.
Every track and artist is fictional; no account or internet is needed.

Genre: 1 Pop  2 Electronic  3 Acoustic
Choose genre (1-3): 2
Mood: 1 Calm  2 Energetic  3 Reflective
Choose mood (1-3): 1
Length: 1 Up to 4 min  2 Over 4 min: 2

Recommendation: Slow Orbit by Rin Echo (6 min)
A spacious instrumental with patient textures.
Why: It matches your Electronic choice, Calm mood, and length preference.
Find another? 1 Yes  2 No: 1

Genre: 1 Pop  2 Electronic  3 Acoustic
Choose genre (1-3): 1
Mood: 1 Calm  2 Energetic  3 Reflective
Choose mood (1-3): 3
Length: 1 Up to 4 min  2 Over 4 min: 1

Recommendation: City Lanterns by Mira Vale (3 min)
Soft vocals for a quiet evening.
Why: It matches your Pop choice, and length preference.
Find another? 1 Yes  2 No: 2
Thanks for exploring.
```

Figure 3: Interactive output and typed choices; complete transcript in `assets/program_output.txt`.

```
MUSIC DISCOVERY ASSISTANT
An offline, rule-based demonstration of digital music discovery.
Every track and artist is fictional; no account or internet is needed.

Genre: 1 Pop  2 Electronic  3 Acoustic
Choose genre (1-3): 1abc
Please enter a whole number from 1 to 3.
Choose genre (1-3): 1.5
Please enter a whole number from 1 to 3.
Choose genre (1-3): 1
```

Figure 4: Real rejection of `1abc` and `1.5`; complete transcript in `assets/invalid_output.txt`.

Part 2 Testing and results

35 black-box checks passed: 18 preference combinations, 8 invalid inputs, 4 EOF cases, repeat, whitespace, menu bounds, fallback and mood priority.

The Python driver executes the compiled programme as a separate process and writes all 35 individual outcomes to `assets/test_results.csv`. These checks validate deterministic behaviour rather than listener satisfaction.

Case	Input	Expected	Actual	Status
C01	1, 1, 1; exit 2	City Lanterns	City Lanterns	PASS
C08	2, 1, 2; exit 2	Slow Orbit	Slow Orbit	PASS
C17	3, 3, 1; exit 2	Paper Harbour	Paper Harbour	PASS
I01	'wrong'; then 1,3,1,2	Reject and recover	Rejected and recovered	PASS
I05	'1abc'; then 1,3,1,2	Reject and recover	Rejected and recovered	PASS
I06	'1.5'; then 1,3,1,2	Reject and recover	Rejected and recovered	PASS
I07	"; then 1,3,1,2	Reject and recover	Rejected and recovered	PASS
E01	"	Clean EOF exit	Clean EOF exit	PASS
R01	2,1,2,1; 1,2,1,2	Two recommendations and normal exit	Two recommendations and normal exit	PASS
B01	invalid length 3 and repeat 7	Reject at both menus and recover	Rejected at both menus	PASS
F01	Pop, Reflective, short	Fallback does not claim a mood match	No false mood match	PASS
P01	Acoustic, Energetic, short	Mood priority; no false length claim	Storm Window; no false length claim	PASS

Run `sh test.sh`. Python is needed to reproduce tests, not to run the C++ programme.

Part 2 Git development record

This screenshot is generated from the actual Git repository. Earlier Netflix commits remain in its history; subsequent Spotify work is honestly recorded as a later AI-assisted revision, not attributed to students who did not make those commits.

```
* 214b90c Revise Spotify Clayton graph and explain trajectories
* 6a967ac Refresh final Spotify report with verified Git snapshot
* a8d8817 Rebuild Spotify Clayton poster report and program evidence
* a3a6704 Stage Spotify music assistant pending Topic 6A graph review
* 73ddf6a Align Netflix graph and customer mapping with supplied Clayton
template
* 620b5d1 Package reviewed report poster and reproducible evidence
* ff2cf6b Document verified Netflix lifecycle model and six-member submission
materials
* 27ea971 Validate complete menu lines and verify recommendation and recovery
scenarios
* a4ee86f Clarify foothold stage and add presentation guidance and disclosure
* 41caa9c Lay out A3 poster and report with readable code capture
* f31ad38 Capture actual program run and local development history
* af73f2b Handle invalid and ended input; add five scenario checks
* 0270ef4 Implement menu inputs and fictional film catalogue
* eddae02 Document scope and learning goals for the movie assistant
```

Figure 5: Real git log --oneline --graph --all. Names and timestamps are in `assets/git_history_authors.txt`. The portable repository bundle is `development_history.bundle`.

Presentation links and assistance disclosure

Source code and Git history access link:

OneDrive video recording: [\[Present Video\]](#)

Source code: [\[Source Code\]](#)

Git history: [\[Git History\]](#)

Speaker	Suggested subject	Approximate time
Vincent Lock Chun Kit	Case and thesis	1:40
Ho Ming Hao	Graph and customer levels	2:20
Soo Kian Rong	Timeline and market impact	2:15
Wong Kee Yuan	Programme design	2:25
Aziel Tan Zheng Chuan	Demo and validation	3:00
See Wing Kit	Testing, critique and close	2:30